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A Neural Network Optimization Approach for Anemia Prediction Using Image Pixels and Hemoglobin Data

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***Abstract*— This study developed a neural network optimization model to predict anemia using image pixel data and hemoglobin levels. The model was designed to address class imbalance and achieved strong performance, with a test accuracy of 97%, precision of 97%, and recall of 96.88%. Data preprocessing techniques such as z-score normalization and ordinal encoding were applied to standardize the input features, while L2 regularization was used to prevent overfitting. The model's performance was assessed using multiple metrics, such as the F1-score and AUC, highlighting its reliability in anemia prediction. The results indicate that this model offers a reliable and efficient approach for non-invasive anemia detection, making it a valuable tool for clinical use in diagnosing anemia based on image data and hemoglobin levels.**

Keywords—Neural network, Anemia prediction, Image pixel data, Hemoglobin levels, Non-invasive detection, medical diagnostics.

I. INTRODUCTION

Anemia, characterized by low hemoglobin levels, remains a critical public health challenge, particularly in regions with limited healthcare resources. Traditional diagnostic methods for anemia, like blood tests, are invasive, time-intensive, and costly, often proving impractical in low-resource settings. Emerging methodologies leveraging neural networks offer an innovative approach for non-invasive anemia prediction by analyzing image-based data and hemoglobin levels. However, existing models often face challenges such as class imbalance, computational limitations, and varied accuracy in different demographic groups. This study aims to address these challenges by developing a neural network optimization model, enhancing prediction accuracy, computational efficiency, and robustness, making it suitable for use in regions with limited healthcare infrastructure [1].

In recent years, machine learning approaches, particularly artificial neural networks (ANNs), have emerged as promising alternatives for non-invasive anemia detection. These techniques can analyze digital image data, such as pixel intensities from images of human skin or mucosa, to predict hemoglobin levels and classify individuals as anemic or non-anemic [2]. Such approaches not only reduce the need for invasive blood tests but also offer the potential for real-time, cost-effective screening in remote and underserved areas.

One promising method is the application of neural networks to analyze the color distribution in images of human tissues, such as the lip mucosa or palm, to estimate hemoglobin levels. Studies using convolutional neural networks (CNNs) to classify anemia based on image data have reported accuracies exceeding 96% [3]. Moreover, smartphone cameras and other low-cost imaging tools have been integrated with machine learning algorithms to provide scalable and accessible solutions for anemia [4].

However, despite these advancements, one of the primary challenges in implementing AI-based anemia detection models in low-resource settings is ensuring that these models are cost-sensitive. Balancing accuracy with the computational and financial costs of deployment is crucial to making these methods widely applicable. In this context, neural networks provide a viable solution by optimizing the prediction process to minimize costs while maintaining high accuracy.

This study aims to develop a neural network optimization approach for anemia prediction, utilizing both image pixel data (red, green, and blue pixel percentages from digital images) and hemoglobin data. By integrating these two sources of information, the model can achieve robust predictions while considering the financial and computational constraints, making it a suitable tool for use in low-resource healthcare settings. This dual-input method offers substantial potential to enhance both the accessibility and efficiency of anemia screening.

II. LITERATUR REVIEW

Recent research has shown significant advancements in using neural networks for non-invasive anemia detection, especially through analyzing image data from areas such as the palm, lip mucosa, and eye conjunctiva. One notable study used Convolutional Neural Networks (CNNs) were employed to classify anemia using palm images, attaining an impressive accuracy of 96.43% [3]. CNNs, with their strong feature extraction capabilities, are particularly

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effective for capturing fine-grained color distributions indicative of anemia. However, their performance relies heavily on the quality and quantity of labeled image data, which can limit their applicability in resource-constrained settings.

In addition, CNN models have been applied in pediatric healthcare to predict early anemia in premature infants based on clinical and test data, showcasing the broader utility of neural networks in health diagnostics [5]. A comparative analysis of CNN models for anemia detection using palm images found that the CNN AllNet model achieved a remarkable accuracy of 96.8%, outperforming other models [6]. This comparison underscores the need for customized architectures to handle subtle feature distinctions in anemia-related image data, though such models can become computationally intensive and may struggle with generalization across different demographics.

Transfer learning, a technique that leverages pre-trained models, has further enhanced the accuracy of anemia detection by utilizing knowledge from related tasks. For instance, transfer learning models applied to lip mucosa images achieved a remarkable accuracy of 99.28% with the Xception architecture [7]. While transfer learning can improve model performance, particularly with limited data, it is sensitive to the choice of the pre-trained model and may not always adapt well to new datasets, potentially leading to overfitting or decreased accuracy on specific features related to anemia.

Other innovative approaches include utilizing deep learning to predict anemia from facial images in emergency scenarios, achieving an accuracy of 84.02% in patient-level predictions [8]. While these models demonstrate potential in real-time anemia detection, their relatively lower accuracy highlights challenges in feature extraction from less constrained, real-world images, where lighting and angle variations can impact prediction quality. Another study applied deep learning to predict hemoglobin concentration from ultra-wide-field fundus images, achieving a mean absolute error of 0.83 g/dl, suggesting that neural networks can predict anemia from non-traditional image types with high precision [9].

Other researchers applied CNNs to palpebral conjunctiva images, achieving an impressive accuracy of 97.94% with ResNet-50 architecture, further supporting CNNs' utility for anemia detection across a range of image inputs [10]. However, the computational demands of ResNet-50 and similar architectures can make deployment challenging in low-resource settings, necessitating optimization strategies or lightweight architectures for practical use.

In peripheral blood smear analysis, a three-tier deep neural network model has been applied to classify anemia severity, with validation accuracies as high as 88.85% [11]. Although effective, these models often require high-quality, magnified blood smear images, limiting their use to controlled lab environments. Additionally, optimization-driven frameworks, such as the whale optimization algorithm applied to CNNs for blood smear images, have demonstrated improvements in both accuracy and training efficiency, showing that neural network optimization techniques can address computational costs [12].

In summary, while CNNs, transfer learning, and other neural network techniques have achieved high accuracy in anemia detection across various datasets, they each come with trade-offs. CNNs provide powerful feature extraction but require large, labeled datasets and computational resources. Transfer learning boosts performance on smaller datasets but can be limited by data-specific nuances, while optimization algorithms show promise in reducing computational burdens. The diverse approaches underscore the potential for neural network optimization to achieve efficient, accurate anemia screening suited to different healthcare environments, especially in low-resource settings.

III. PROPOSED METHOD

This research proposes a novel neural network optimization model for predicting anemia by integrating hemoglobin levels and color pixel data derived from medical images. The primary goal of this method is to enhance prediction accuracy, particularly for the anemic class, by addressing class imbalance.

A. Dataset

The dataset utilized in this study was specifically curated to support the development of machine learning models for anemia prediction, integrating both image pixel data and hemoglobin levels [13]. It consists of a total of 104 cases, with each entry representing an individual case containing various features relevant to anemia prediction. Of the 104 cases, 26 are classified as anemic (25%), while 78 are classified as non-anemic (75%), demonstrating a class imbalance that could potentially impact model performance. Each case in the dataset includes a unique identifier, the sex of the individual. Additionally, the dataset included the percentage distribution of red, green, and blue pixels extracted from the associated images. These color distribution features provide valuable insights, as they reflect physiological changes potentially associated with hemoglobin variations related to anemia.

TABLE I. DETAIL DATASET

Data	**Attributes**	**Class**	**Distribution Class**		**Intances**
			Yes	***No***	
Anemic	18	2	26	78	104

Additionally, each entry provides the individual's hemoglobin (Hb) level, measured in grams per deciliter (g/dL), which serves as a crucial clinical biomarker in diagnosing anemia. The anemia status of each individual is noted under the 'Anaemic' column, with a categorical label indicating either "Yes" for anemic or "No" for non-anemic cases [14]. For the purposes of machine learning, this categorical variable is converted to a binary form, assigning a value of 1 for anemic cases and 0 for non-anemic cases. To address the dataset's class imbalance, no specialized handling techniques, such as undersampling, synthetic data generation or oversampling, were applied. Instead, the model relied on L2 regularization and carefully selected evaluation metrics, including precision, recall, and F1-score, to minimize the effects of imbalance on its performance. By focusing on these performance metrics, the model could maintain high predictive accuracy while controlling weight adjustments, thus ensuring balanced sensitivity across both classes. The dataset's integration of image-based pixel data and

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hemoglobin levels provides a robust feature set, enabling the neural network model to effectively learn patterns associated with anemia, while remaining suitable for cost-efficient and non-invasive diagnostic applications.

B. Data Preprocessing

Data preprocessing is critical for this model, as it directly affects the neural network's ability to learn from diverse input features. Z-score normalization was selected over other scaling methods due to its ability to standardize input features with differing scales [15], such as pixel intensities and hemoglobin levels, bringing each feature to a similar range with a mean of 0 and standard deviation of 1. This ensures uniform feature scaling, which enhances convergence speed and minimizes the risk of model bias toward larger values. Ordinal encoding was used to transform categorical features, such as sex, into numeric form, which prevents the model from incorrectly assigning hierarchical importance to these values.

TABLE II. SCALING TRAIN TEST DATASET ANEMIC

Class	Train Distribution		Test Distribution	
	Instances	*(%)*	*Instances*	*(%)*
No	54	75.00%	24	75.00%
Yes	18	75.00%	8	75.00%

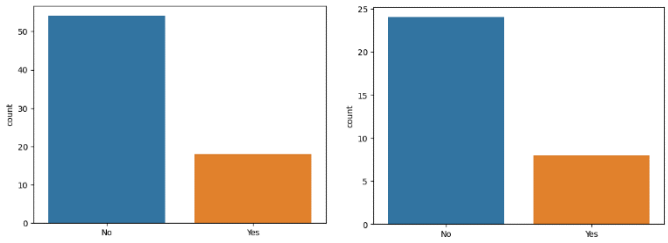


Fig. 1. Visualization of the anemic dataset after scaling

C. Neural Network

The neural network model includes an input layer designed to process hemoglobin data and RGB pixel percentages. This is followed by three hidden layers comprising 256, 512, and 1024 neurons, respectively. Each hidden layer utilizes the ReLU activation function to introduce non-linearity, essential for capturing intricate patterns in the data. To prevent overfitting, dropout regularization with a rate of 0.2 is applied to each hidden layer. The model leverages the Adam optimizer, known for its efficient convergence in neural networks, with a learning rate of 0.0001. Hyperparameters such as layer dimensions, learning rate, and dropout rates were meticulously fine-tuned through cross-validation to ensure optimal performance.

In an Artificial Neural Network (ANN), each neuron calculates a weighted sum of its inputs, incorporates a bias term, and processes the result using an activation function [16].

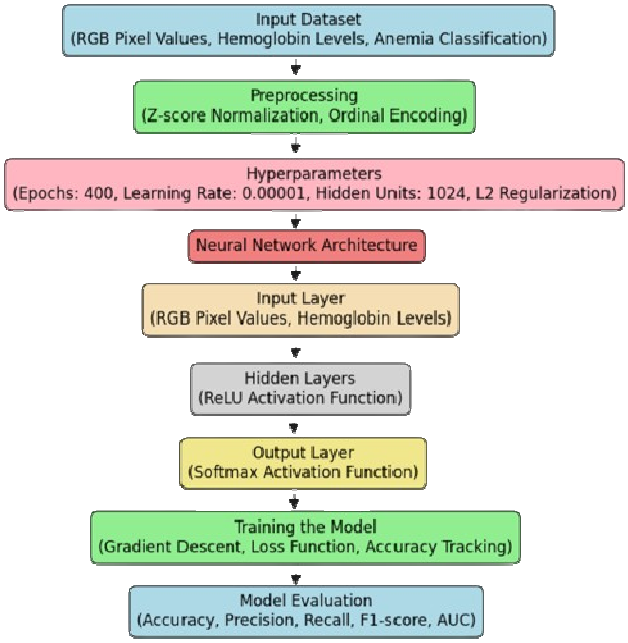


Fig. 2. Neural network architecture

For the model setup described (with two dense layers), we can express this mathematically as follows:

Step-1. Input Layer: Dense Layer with ReLU Activation

The first layer computes the output for each neuron as:

$$Z^{(1)} = W^{(1)}X + b^{(1)} \quad (1)$$

The first dense layer of the neural network computes an output $Z^{(1)}$ before applying the activation function. This output is derived using the equation (1), where $W^{(1)}$ is the weight matrix of the first layer with dimensions $D \times H$. Here D represents the number of input features, and H denotes the number of hidden units in the first layer. The input data, X, has dimensions D, and the bias term, $b^{(1)}$, is a vector with dimensions H. This computation forms the basis for the subsequent activation and learning process in the network.

The output of the dense layer is then passed through the ReLU activation function, adding non-linearity to the model:

$$A^{(1)} = ReLU(Z^{(1)} = \max(0, + Z^{(1)}) \quad (2)$$

Where, $A^{(1)}$ is the activation of the first layer after applying ReLU. ReLU, short for Rectified Linear Unit, is an activation function that outputs the input value if it is positive and returns 0 for any negative input. [17].

Step-2. Output Layer: Dense Layer with Softmax Activation

The second layer computes the output for the final classification:

$$Z^{(2)} = W^{(2)}A^{(1)} + b^{(2)} \quad (3)$$

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The output of the second dense layer, $Z^{(2)}$, is computed before applying the activation function. This calculation uses the equation (3), where $W^{(2)}$ represents the weight matrix of the second layer with dimensions $H \times K$. Here H is the number of hidden units in the first layer, and K is the number of output classes. $A^{(1)}$ is the activation output from the first layer, and $b^{(2)}$ denotes the bias term for the second layer with dimensions K.

Then, the Softmax activation function is applied to the output:

$$\hat{y}_i = \frac{exp(Z_i^{(2)})}{\sum_{j=1}^{K} exp Z_j^{(2)}} \quad (4)$$

Here, $\hat{y}_i$ denotes the predicted probability for class i, derived from $Z_i^{(2)}$, the second layer's output. The Softmax function normalizes these outputs into probabilities that sum to 1.

Step-3. Regularization (L2 Regularization)

To mitigate overfitting, L2 regularization, also referred to as ridge regularization, is applied [18]. The L2 regularization term is incorporated into the cost function.

$$J_{total} = J + \frac{\lambda}{2m}^{(1)} = \sum W^{(1)2} + \sum W^{(2)2} \quad (5)$$

Where, J_{total} is the total cost (including the regularization term). Here, J represents the original cost function, such as cross-entropy loss. The parameter λ controls the degree of regularization applied, while m denotes the number of training examples. The terms $\Sigma W^{(1)2}$ and $\Sigma W^{(2)2}$ refer to the sum of the squared weights in the first and second layers, respectively, which contribute to the L2 regularization penalty.

D. Training the Model

In the training phase of the neural network, Gradient Descent is employed to optimize the model by minimizing the error through Backpropagation [19]. During each epoch, the process begins with a forward pass, where the model computes the output probabilities for the training data. The error, or loss, is calculated using a cost function that includes an L2 regularization term to prevent overfitting by penalizing large weights.

After computing the loss, the backward pass adjusts the model's weights and biases to reduce the error. This is accomplished by adjusting the parameters in the direction that reduces the cost, using the gradient of the loss function as a guide. During training, the model's performance is continuously assessed on both the training and testing datasets. The accuracy of predictions, as well as the loss, is recorded at each epoch, allowing the model's learning progress to be monitored. This process continues iteratively until the model converges or completes the specified number of epochs.

E. Model Evaluation and Hyperparameter

The model's performance is evaluated at regular intervals (every 10 epochs) by printing the loss and accuracy for both training and testing sets. After training, final evaluation on the test set is conducted, where metrics such as Accuracy, F1 Score, Precision, and Recall, and ROC AUC are calculated. Results are visualized using a Confusion Matrix and Precision-Recall Curve, with per-class metrics for deeper insights into performance.

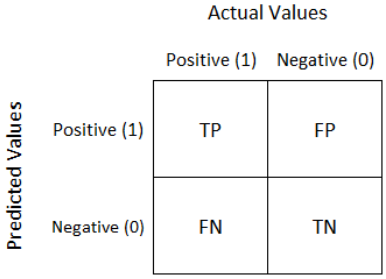


Fig. 3. Confusion matrix illustration

The confusion matrix summarizes classification outcomes, with its components representing different results. True Positives (TP) denote correctly identified positive instances, False Positives (FP) reflect negative instances misclassified as positive, True Negatives (TN) indicate correctly classified negative instances, and False Negatives (FN) represent positive instances wrongly classified as negative. By examining the confusion matrix, the model's classification performance can be evaluated. This study uses metrics such as Accuracy, Precision, Recall, F1-Score, and ROC AUC to assess the effectiveness of the classification models, offering comprehensive insights into their accuracy and reliability in data classification [20].

The ROC AUC Score is another valuable metric for evaluating the performance of a multiclass logistic model, particularly in scenarios with imbalanced class distributions. It offers a robust measure of the model's effectiveness in multiclass classification tasks [21].

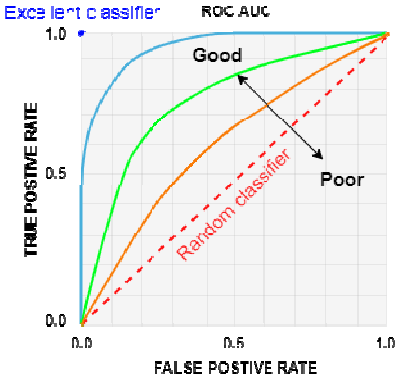


Fig. 4. ROC curve and auc illustration

IV. RESULTS AND DISCUSSION

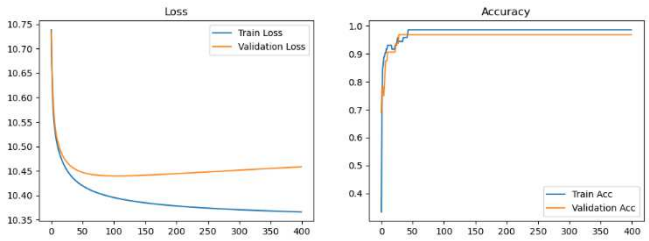
The neural network model was evaluated on an anemia dataset using a personal computer equipped with an Intel Core i5 processor, 4 GB of RAM, running Windows 10. The implementation was carried out in Python using Jupyter Notebook. The model's performance was assessed across both training and testing sets, with a primary focus on handling imbalanced medical data without explicit class balancing techniques like oversampling or undersampling. The model consistently improved over 400 training epochs,

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achieving a final training accuracy of 99% and a test accuracy of 97%, indicating effective generalization to unseen data. The final training and test loss values of 10.3662 and 10.4583, respectively, reflect minimal overfitting, while the accuracy plateauing around Epoch 49 demonstrates efficient convergence.

Given the dataset's class imbalance (25% anemic and 75% non-anemic cases), performance metrics such as precision, recall, F1-score, and AUC were used to assess sensitivity and specificity across both classes. For non-anemic cases, the model achieved 100% recall and 96% precision, resulting in an F1 score of 0.98, signifying a strong balance between precision and recall. For anemic cases, the model achieved 87.5% recall and 100% precision, with an F1 score of 0.933, demonstrating effective performance in the minority class. The AUC of 91% further supports the model's robustness in distinguishing between anemic and non-anemic cases.

These high metrics suggest that the model is both sensitive and specific in its predictions, maintaining strong classification performance despite the imbalanced dataset. The use of image pixel data and hemoglobin levels as features, alongside careful preprocessing and L2 regularization, allows the model to accurately predict anemia without class balancing, making it suitable for real-world clinical applications.



(a) (b)

Fig. 5. The classification results for the anemic using the neural network model (a) the curves of loss (per epoch) and (b) the curves of accuracy (per epoch)

The neural network model trained on the anemia dataset demonstrated high performance across multiple metrics, underscoring its effectiveness in predicting anemia using image pixel data and hemoglobin levels. Over 400 epochs, the model consistently improved, achieving a final training accuracy of 99% and a test accuracy of 97%, indicating strong generalization to unseen data. The minimal difference between final training and test loss values, 10.3662 and 10.4583 respectively, reflects limited overfitting and effective learning from the dataset, with accuracy plateauing around Epoch 49, signaling efficient convergence.

Comprehensive performance metrics, including F1-score, precision, and recall, highlighted the model's ability to handle class imbalance, confirming it as both sensitive and specific in its predictions. The confusion matrix further demonstrated accurate classification, with the model achieving 100% recall and 96% precision for non-anemic cases (F1 score of 0.98) and 87.5% recall and 100% precision for anemic cases (F1 score of 0.933). These results suggest slightly better performance in non-anemic predictions but also reflect excellent precision in detecting anemia.

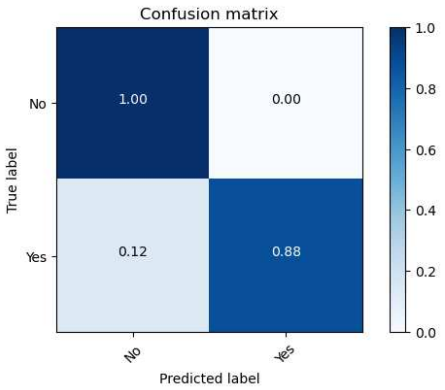


Fig. 6. Confusion matrix results

TABLE III. MODEL PERFORMANCE PER CLASS

Class	Accuracy	Recall	Precision	F1-score	AUC
No	0.969	1.000	0.960	0.980	0.910
Yes	0.969	0.875	1.000	0.933	0.910

To further understand the model's performance, an error analysis was conducted to examine false positives and false negatives. This analysis revealed that while the model maintains high sensitivity, especially for non-anemic cases, the one false negative in the anemic class indicates a potential area for improvement. Additional training data or refined feature selection could help enhance sensitivity for challenging cases, potentially by identifying subtle variations in color distribution linked to hemoglobin levels.

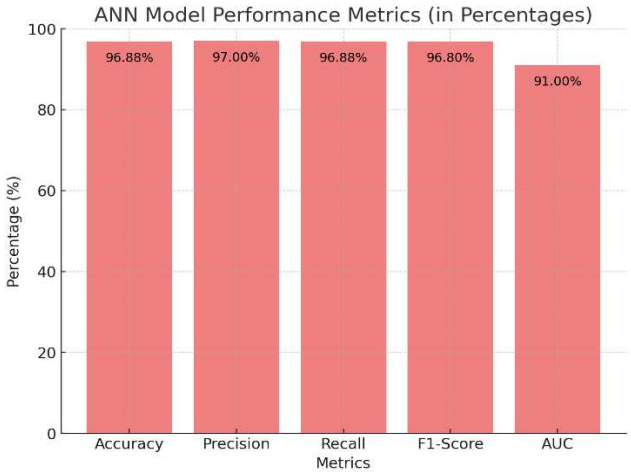


Fig. 7. Neural network optimization performance for anemic datasets

The ANN model demonstrates strong performance, with accuracy of 96.88%, precision of 97.00%, and recall of 96.88%, indicating that it accurately predicts anemia while minimizing both false positives and false negatives. The F1-score of 96.80% demonstrates balanced precision and recall, while the AUC of 91.00% indicates the model's strong capability to differentiate between anemic and non-anemic cases. Overall, the model is highly effective for anemia prediction.

V. CONCLUSION

The neural network model developed in this study effectively addresses the challenges associated with non-invasive anemia detection, achieving high accuracy and

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computational efficiency without the need for class balancing techniques. By integrating image pixel data with hemoglobin levels, the model demonstrates strong predictive capabilities. This makes it especially suitable for low-resource settings where conventional diagnostic tools are scarce. The model achieved a high test accuracy of 97%, with a precision of 97% and a recall of 96.88%, highlighting its robust performance in handling imbalanced data.

The data preprocessing steps, including z-score normalization and regularization, played a critical role in enhancing model reliability and reducing potential bias. This robust framework indicates that neural networks, when carefully optimized, can be reliable tools for non-invasive anemia screening in diverse healthcare environments.

Future research could build on these findings by expanding the dataset to improve generalizability, incorporating additional clinical features to refine predictions, and developing real-time applications for on-site diagnostics. Moreover, investigating alternative neural network architectures and implementing advanced techniques to handle class imbalances could further improve the model's performance, expanding its potential in clinical diagnostics and supporting more accessible healthcare solutions.

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